

SIH 2026 — Entity-Wise Solution Framework

Problem Statement ID: 26.036

Development of an Online Verification System for Weighing & Measuring Instruments

Digitalizing and automating the verification workflow, calculations, records, scheduling, certification, and consumer verification under the Legal Metrology framework.

Entity	Primary Solution Focus
Instrument Owners & Traders	Digital applications, readiness checks, scheduling, alerts, fees
Legal Metrology Officers (LMOs)	Smart scheduling, unified records, digital inspection, calculations
Directorate & System Administrators	Central dashboard, certificates, auditability, enforcement
General Consumers	QR verification, search, complaints, complaint tracking
Manufacturers & Importers	Model approval, compliance, bulk registration, initial verification
GATCs & Licensed Repairers	Digital testing, reporting, coordination, audit-ready records

1. Instrument Owners & Traders

Key Problems

- Complex paper-based applications and physical visits to government offices.
- Missed verification/re-verification deadlines.
- Limited transparency regarding fees, scheduling, and application status.

Recommended Solutions

Digital Verification Application

- Provide an online application for verification and re-verification.
- Replace paper forms with structured digital forms and document upload.
- Generate a unique application ID for tracking.

Pre-Verification Readiness Check

- Provide instrument-specific requirements and a preparation checklist.
- Allow preliminary readings or checks where appropriate.
- Clearly distinguish preliminary self-checks from official Legal Metrology verification.

Automated Appointment Scheduling & Alerts

- Show available verification slots and allow appointment requests.
- Consider officer availability, workload, location, instrument type, and urgency.
- Send notifications for scheduled inspections and status changes.

Application Tracking

- Track stages such as Submitted, Scrutiny, Scheduled, Inspection/Testing, Approved/Rejected, and Certificate Issued.
- Provide notifications whenever the application status changes.

Deadline Tracker

- Store verification validity/expiry dates for registered instruments.
- Send proactive SMS/email/app alerts before expiry.
- Allow early re-verification requests to reduce accidental non-compliance.

Transparent Standardized Fees

- Calculate applicable fees automatically according to configured legal fee rules.
- Display the fee before payment.
- Provide online payment and digital receipt generation.

2. Legal Metrology Officers (LMOs)

Key Problems

- Inefficient scheduling under high workload and limited staff.
- Duplicate records caused by inconsistent business/entity information.
- Gaps in access to previous equipment history.
- Repeated data entry across separate systems.

Recommended Solutions

Area-Based Smart Scheduling

- Group appointments by locality or geographical area.
- Schedule nearby inspections together to reduce unnecessary travel.
- Consider officer availability, workload, instrument type, and priority.

Officer Workload Dashboard

- Display pending applications, today's inspections, upcoming appointments, and overdue cases.
- Provide area-wise workload visibility for better allocation.

Unified Instrument Record

- Create a unique digital identity for each instrument.
- Maintain serial number, manufacturer, model, owner, verification status, previous verification, and inspection history.

GST-Based Entity Tracking

- Use GSTIN where applicable as a business/entity identifier to reduce duplicate business records.
- Keep the instrument's own unique serial/digital identity separate from the business identifier.

Digital Inspection & Verification Form

- Allow LMOs to record inspection and test results digitally.
- Use mandatory fields and validation to reduce incomplete records and transcription errors.

Automated Verification Calculation Engine

- Perform applicable verification calculations from entered test readings.
- Reduce manual calculation errors.
- Store inputs, calculated results, and final outcomes for traceability.

Digital Certificate Generation

- Generate the digital certificate after successful verification and approval.
- Link the certificate directly to the instrument's digital record.

Backend System Integration

- Where government systems permit, integrate licensing, payment, e-stamping, and related portals through APIs/backend services.
- Reduce repeated entry of identical trader and instrument information.

Rule-Based Record Retention

- Configure record-retention periods according to applicable government policies and instrument-specific requirements.
- Keep older records accessible for authorized audits and enforcement when required.

3. Directorate & System Administrators

Key Problems

- No real-time centralized view of verification and enforcement activity.
- Traceability gaps in actions and records.
- Risk of fraudulent or forged certificates.
- Difficulty monitoring complaints and compliance at scale.

Recommended Solutions

Real-Time Centralized Dashboard

- Monitor applications, pending/completed verifications, expired certificates, complaints, officer workload, and GATC activity.
- Provide state, district, and area-wise filtering.

Digital Certificate Repository

- Maintain official digital certificates centrally.
- Allow authorized stakeholders to retrieve certificate records.
- Track certificate status such as valid, expired, suspended, or cancelled.

QR-Based Certificate Authentication

- Generate a unique QR code linked to the official instrument/certificate record.
- Verify certificate authenticity against the central system.
- Display certificate status and issuing authority without exposing unnecessary private information.

Tamper-Evident Audit Trail

- Record timestamps and authorized users for application, inspection, result entry, certificate issuance, modification, suspension, and cancellation.
- Support accountability and audit investigations.

Automated Complaint Cross-Checking

- Cross-check the instrument's verification status when a consumer submits a complaint.
- Flag expired, unverified, suspended, or otherwise relevant records for authority attention.

Enforcement Analytics

- Identify areas with high numbers of expired instruments, failed inspections, complaints, or repeat violations.
- Help authorities prioritize inspections and enforcement activity.

Role-Based Access Control

- Provide separate permissions for LMOs, GATCs, administrators, Directorate users, and owners/traders.
- Prevent unauthorized modification of official records.

4. General Consumer

Key Problems

- Consumers cannot easily verify whether an instrument is officially verified.
- Reporting suspected irregularities is inconvenient.
- Limited visibility into complaint status.

Recommended Solutions

Instrument QR Code

- Provide a unique QR code for each verified instrument where applicable.
- Allow consumers to scan the code using a smartphone and access official verification information.

Instant Verification Status

- Display instrument identification, verification status, validity/expiry, verification authority, and certificate/reference number.
- Show only the information necessary for public verification.

Manual Unique-ID / Serial Search

- Allow consumers to manually enter the instrument's unique identification or serial number when QR scanning is unavailable.

Direct Complaint Portal

- Provide a complaint option directly from the instrument verification page.
- Allow submission of a description and supporting evidence such as photographs where permitted.

Complaint Tracking

- Generate a complaint/reference number.
- Allow consumers to track submitted, under-review, assigned, action-taken, and resolved/closed stages.

Complaint-to-Instrument Linking

- Automatically associate complaints with the relevant instrument record.
- Help authorities identify repeated complaints associated with the same instrument or business.

5. Manufacturers & Importers

Key Problems

- Model approval bottlenecks.
- Complex multi-state compliance.
- Delays in bulk initial stamping and verification.

Recommended Solutions

Online Model Approval Application

- Submit model approval applications digitally.
- Upload technical documents and required certificates.
- Track the approval workflow online.

Model Approval Tracking

- Show submitted, pending scrutiny, query raised, approved, and rejected stages.
- Notify applicants when corrections or additional documents are required.

Centralized Model Registry

- Maintain approved model information digitally.
- Link instruments manufactured/imported under an approved model to the relevant model record.

Multi-State Compliance Dashboard

- Provide a consolidated view of applicable licensing and compliance requirements by state.
- Track submissions, approvals, and pending actions.

Bulk Instrument Registration

- Allow manufacturers/importers to register multiple instruments through structured bulk upload or APIs.
- Generate individual instrument records from validated batch data.

Batch Initial Verification/Stamping Workflow

- Create batches of instruments for initial verification/stamping.
- Track batch status and assigned authority/test centre.
- Reduce repetitive manual coordination.

Digital Notifications

- Notify manufacturers/importers about missing documents, scheduled inspections, testing completion, approval, rejection, or correction requirements.

6. GATCs & Licensed Repairers

Key Problems

- Delayed official stamping and verification coordination.
- Redundant testing intervals.
- Manual reporting and record vulnerability.
- Difficulty producing records during audits.
- Need to distinguish authorized providers from unlicensed competition.

Recommended Solutions

Digital Test-Centre Workflow

- Receive assigned testing/verification requests digitally.
- View instrument details and required test parameters before testing.
- Update test status in real time.

Standardized Digital Test Reports

- Record test readings through instrument-specific digital forms.
- Reduce handwritten reports and transcription errors.
- Capture required fields consistently.

Digital Test Data Repository

- Store historical test results against the instrument's unique ID.
- Allow authorized officers to retrieve previous test data.

LMO–GATC Coordination

- Notify LMOs when testing is completed.
- Allow authorized officers to review results before official certification/stamping.
- Track pending approvals and actions.

Automated Test Calculations

- Perform predefined calculations from recorded test readings.
- Reduce manual calculation errors.
- Preserve raw readings alongside calculated results.

Audit-Ready Records

- Maintain timestamps, tester identity, instrument identity, and test results.
- Enable rapid retrieval of records during inspections and audits.

Licensed Service Provider Registry

- Maintain a digital registry of authorized GATCs and licensed repairers.
- Allow authorized users and, where appropriate, the public to verify provider status.

Complete Instrument Test & Repair History

- Link testing and authorized repair records to the instrument's unique identity.
- Create a traceable lifecycle history for the instrument.

Overall Solution Concept

One Instrument → One Digital Identity → Complete Lifecycle

- Every instrument receives a unique digital record connected to its owner/trader, manufacturer/importer, verification history, testing data, certificate, expiry date, QR verification, complaints, and relevant enforcement records.
- The same underlying instrument record connects the Owner/Trader, LMO, GATC/Licensed Repairer, Manufacturer/Importer, Directorate/Admin, and Consumer workflows.
- The core end-to-end workflow can be demonstrated as: Digital Application → Scheduling → Inspection/Testing → Automated Calculations → Approval → Digital Certificate → QR Verification → Consumer Complaint → Authority Action.
- The platform should use rule-based/configurable verification parameters for different instrument categories so that calculations and validation can evolve without redesigning the complete system.

SIH Implementation Priority

- Must Have: Digital application, instrument digital identity, LMO workflow, verification calculations, digital certificate, QR verification, complaint tracking, and central dashboard.
- High-Value Innovation: Area-based smart scheduling, configurable verification rules, complete instrument lifecycle timeline, and integrated complaint-to-enforcement workflow.
- Future Scope: Expanded government-portal integrations, advanced analytics, broader GATC/repairer workflows, and additional automation.

ONE INSTRUMENT • ONE DIGITAL IDENTITY • COMPLETE LIFECYCLE

Interconnected Workflow of Online Verification System for Weighing & Measuring Instruments

